

Video transcript

Physical threats

When you think about cybersecurity threats, does your mind automatically think of digital threats? Threats can be more direct than that. Physical security safeguards computer hardware and other physical infrastructure. It defends assets from direct threats, such as theft, damage, or unauthorized access. No firewall or antivirus software can prevent someone from plugging directly into a network port or stealing a hard disk from a data center. If someone accesses a data center or steals a laptop, they might be able to steal sensitive information, damage hardware, or install malicious software directly.

Physical security takes many forms. Consider the process Maria, a cybersecurity analyst, follows when she gets to work each day. First, she uses her key card to enter the lobby. She waves to the security guard and the receptionist on her way. Recognizing her, they say, "Good morning", then return to their work. Maria continues to the elevator, passing the cameras located in the lobby, hallway, and elevators. She uses her key card to access the floor where the security operations center is located. She is the first to arrive, so the motion detecting lights turn on when she enters. When she gets to her desk, she turns on her computer, enters her password, and verifies her identity with a thumbprint. All these steps help protect physical security by deterring, preventing, or delaying malicious actors from entering by identifying unauthorized people, by defending assets directly, or by providing other functions. The exact configuration of controls might vary, but they all serve a purpose, and they are layered so that if one fails, another is in place to compensate. This is just the tip of the iceberg when it comes to physical threats.

Much of the world's critical infrastructure is now networked, which means greater potential for physical breaches. Utility companies, transportation companies, and government agencies are prime examples of organizations that must take physical security very seriously. Environmental threats also can harm physical equipment. Extreme temperatures, humidity, fire, and moisture can destroy computer hardware. So can electrical surges or electromagnetic interference. Organizations need to plan defenses for these threats.

Companies can install detective controls to detect unusual conditions that may pose a threat. If dangerous conditions are detected, corrective controls, like air conditioning switches, can take action to remove the threat. For example, a thermostat can detect a high temperature and trigger air conditioning

switches to lower the temperature. It's easy to forget that digital systems and information rely on physical components. For effective cybersecurity, though, it's crucial to keep computers, storage devices, wiring, and other physical assets safe and secure.